

COVID-19 International Spread Chain Analysis: A Systematic Literature Review

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Abstract

The fast spread of COVID-19 emphasized the need to understand transmission pathways and international disease dissemination. This study presents a PRISMA-based systematic literature review of 46 papers related to data mining, network analysis, machine learning, and spatiotemporal modeling for COVID-19 international spread analysis. The selected studies were analyzed with respect to their methodologies, datasets, and contributions to three research questions concerning travel-origin countries, transmission patterns, and computational approaches for identifying potential spread routes. The results indicate that temporal, spatial, and network transmission patterns are well studied, whereas international origin–destination relationships remain relatively underexplored. Finally, this review identifies significant research gaps and gives a comprehensive synthesis of analytical approaches for investigating international COVID-19 spread.

Keywords: COVID-19 International Spread, Transmission Pathway, Mobility Analysis, Network Analysis, Spatiotemporal Analysis, Sequence Mining, Association Rule Mining, Spread Prediction

1 Introduction

The speedy worldwide transmission of Coronavirus Disease 2019 (COVID-19) showed how infectious diseases can spread through interconnected transportation and mobility networks, transforming local outbreaks into global public health emergencies within a

short period. Acknowledging international spread mechanisms is crucial for epidemiological surveillance and for improving alertness against future pandemics. Though significant studies have carried out cases forecasting, reproduction rates, external factors, and public health interventions, conversely, a few studies have emphasized on understanding how transmission pathways and origin–destination relationships contribute to the international spread of infectious diseases[1, 2, 3]. From a practical perspective, understanding worldwide transmission pathways can assist early warning systems, risk assessment, and health policy aimed at controlling future outbreaks. From a scientific view, COVID-19 has motivated massive research, including machine learning, data mining, network science, spatiotemporal analysis, and statistical modeling [4, 5]. Typical analytical flows include collecting epidemiological data, mobility and surveillance records, preprocessing various datasets, and applying computational techniques to characterize transmission dynamics and spread patterns. Network reconstruction and transmission-tree studies further focus on the importance of demonstrating infections as interconnected pathways among people, cities, regions, and countries [6, 7]. In spite of valuable improvements, several challenges remain. International spread analysis is difficult because of heterogeneous reporting practices, incomplete travel reports, temporal uncertainty, differences in surveillance systems, and the difficulty of integrating diverse analytical approaches into interpretable frameworks. Furthermore, existing research mainly focuses on local outbreaks, incidence prediction, or isolated methodological perspectives, while comparatively few provide a comprehensive understanding of international transmission mechanisms and high-risk pathways[8, 9].

The objectives are to identify major analytical approaches, examine transmission patterns and origin–destination relationships, and highlight research gaps related to future spread pathways. To address these issues in a structured manner, the present review is guided by the following research questions: RQ1: Which travel-origin countries or regions are most frequently associated with international first-case introductions? RQ2: What temporal, spatial, or network transmission patterns emerge among countries from rapid spread evidence? RQ3: What analytical methods and computational approaches provide interpretable evidence for potential future spread routes or high-risk pathways?

The remainder of this paper is organized as follows. Section 2 presents the literature review and summarizes findings from the selected studies. Section 3 describes the PRISMA-based methodology and screening procedures. Section 4 reports the results, contributions, research question matching analysis, and charts. Section 5 provides a comparative analysis with related survey studies. Finally, Sections 6 and 7 discuss the major findings, limitations, and conclusions, together with directions for future research.

2 Literature Review

COVID-19 has generated an unprecedented stream of research, covering epidemiology, machine learning, data mining, network science, and also geo-spatial analytics[10, 11, 12]. As a result, the literature is fragmented across different analytical traditions and

real-world application areas. A systematic literature review is needed to glue together what’s already been found, contrast the methodologies, spot repeated limitations, and decide what the methodological base is for international spread analysis. The early reviews were mainly focusing on forecasting models and machine-learning techniques, but later on, surveys moved toward application-focused angles, including mobility analysis, contact tracing, network reconstruction, privacy-preserving analytics, and spatiotemporal studies [13]. Still, rather than examining interpretable routes between countries and global transmission, most of the existing reviews are centered on case prediction and local transmission dynamics rather than the international transmission mechanisms. As a result, the present review aims to fill that gap by bringing together the literature from the viewpoint of global spread analysis, and also aims to explain transmission pathways using computational approaches. . Section 2.1 gives an overview of the main research articles, while Section 2.2 synthesizes outcomes from review and survey papers [14].

2.1 Insights From Research

Selected 46 papers, which are mostly primary research studies, can be classified in these four major themes based on their similarities in formulation of problems, analytical methods, and evaluation protocols.

2.1.1 Data Mining and Pattern Discovery Studies

Eight papers mainly focus on association-rule mining and pattern discovery techniques [4, 15]. These studies have used diverse datasets, such as clinical symptoms, hospital records, behavioral surveys, social media information, and epidemiological observations. Most of them rely on Apriori, FP-Growth, or frequent item set mining type algorithms, and then they look at the results using support, confidence, and lift. The overall purpose is to find meaningful connections between symptoms, self-protective habits, clinical outcomes, and risk factors.[16, 17, 18] Discovering interpretable relationships among symptoms, self-protective behaviors, clinical outcomes, and risk factors is their main purpose. To improve prediction performance, several studies have integrated pattern discovery methods using forecasting approaches such as ARIMA and ARIMAX [4, 19, 20]. A typical methodological advantage is explainability, whereas limitations include localized datasets, restricted sample sizes, and limited external validation. These studies demonstrate the importance of data-driven methods for revealing hidden relationships and forming explainable insight.

2.1.2 Spatiotemporal and GIS-Based Studies

Twelve papers examine the spread of COVID-19 from a spatial and temporal angle [19, 20]. Papers lean on national tracking records along with mobility datasets, Population context, environmental variables, and regional data. Methods like spatial clustering approaches, geographic information systems, time-series analysis, and statistical modeling are commonly used. [21, 22, 23]. Their reports consistently reveal the influence of mobility, geographic nearness, population density, and socioeconomic

conditions on transmission dynamics. Reviews use correlation checks, prediction accuracy scores, and measures of spatial association for evaluation procedures n[24, 23]. Although insights are valuable for understanding how outbreaks spread across places, most of them emphasize Occurrence rates and local clusters, rather than the international transmission processes. [25, 26]. Finally, limited comparability across datasets, plus regional heterogeneity, ends up being a recurring problem.

2.1.3 Network and Transmission Studies

Ten papers analyze transmission path and network structures[1, 27, 28]. Datasets consist of features like contact tracing notes, transmission trees, mobility graphs, and interactions across cities.[29, 30, 31, 32]. To show infections as interconnected relationships, Graph-based frameworks, network science methods, and reconstruction algorithms are utilized. In evaluation, they usually check network metrics, tracing accuracy, and transmission properties[6, 29, 33]. These studies suggest that propagation is not random; it shows clustering effects and even hierarchical organization. One big strength is that they preserve connectivity, and the temporal progression of disease spreads pretty well. Still, most of them stick to patient scales, community boundaries, or city-level patterns, and rarely explore links between countries. So, even if their approaches are relevant, the Scope stays smaller than the global transmission, which the current study focuses on.

2.1.4 Modeling, Forecasting, and Contextual Studies

The central theme of ten papers is predicting models and the environmental factors that form the pandemic [34, 35, 36, 37]. To clarify differences in transmission and mortality across places, these papers use statistical models, machine learning methods, environmental assessments, and socioeconomic indicators. Their evaluation routines are usually based on prediction accuracy, error measures, and comparative analyses [38, 39, 40, 41, 7]. These studies demonstrate the influence of climate patterns, policy steps, demographic features, and mobility curbs on pandemic evolution. Rather than analyzing how COVID -19 spread between countries, most of the papers are focusing on case numbers and explaining disease occurrences. Another issue is leaning heavily on black-box prediction models without any Interpretation [42, 43, 32]. Across all primary studies, several methodological trends appear. First, most datasets are local rather than international. Second, predictive performance is more highlighted than Interpretability. Third, validation is often performed on local samples without external generalization. Finally, few studies directly examine international transmission or origin-destination links among countries. These findings demonstrate the gap in understandable international spread analysis. As a result, they push toward combining multiple computational ways to make transmission pathways more understandable [42, 43, 32, 40, 41].

Overall, the reviewed literature indicates that COVID-19 analytics has gained from a mix of perspectives: machine learning, data mining, network science, and spatiotemporal analysis. However, studies are mostly focusing on local outbreaks, case forecasting, or single-method angles rather than exploring international spread mechanisms and providing interpretable evidence about transmission pathways across

countries [40, 41, 34, 35, 36]. As a result of these limitations, a systematic review focused on international spread analysis is needed.

2.2 Findings From Systematic Reviews and Surveys

Among the selected papers, six are essentially review or survey studies [44, 8, 9, 45]. Their inclusion criteria are mostly revolving around machine learning and data mining, digital contact tracing, privacy-preserving analytics, and COVID-19 modeling approaches [46, 7]. For most surveys, it comes out that machine learning, deep learning, statistical forecasting, and network analysis have been dominating pandemic research [8, 9]. There are also the pattern discovery-oriented studies, which underline how useful interpretable knowledge extraction is, along with co-occurrence analysis [45, 44, 46]. The privacy-oriented surveys had shown the problems associated with the quality of data, implementation feasibility, and ethical concerns [45, 44].

According to most of the review studies, COVID-19 analytics had been done mostly using multidisciplinary approaches integrating epidemiology, artificial intelligence, and network science. But there is disagreement on how generalizable the models really are, plus reproducibility, calibration, and whether transferability works cleanly across different populations. Also, most surveys point to the lack of explanation and external validation as ongoing challenges [8, 46, 44]. Another frequent worry is that a lot of prediction models chase accuracy but forget about interpretability and the actual practical deployment in real settings. The pursuit of accuracy and the ignoring of practical deployment and interpretability is another frequent concern that is observed.

Compared with earlier reviews, this SLR introduces three important expansions. To begin with, it focuses more on the analysis of international spreads than on the prediction of COVID-19. Next, using systematic screening and thematic grouping, findings from data mining, network science, and spatiotemporal studies are merged in a more consistent way. Finally, it highlights interpretable transmission mechanisms and computational approaches to supplement other surveys that mostly focus on forecasting performance or clinical applications.

3 Methodology

This paper follows a Systematic Literature Review (SLR) methodology the Meta-Analyses (PRISMA 2020) framework. The methodology consists of four major phases: identification, screening, eligibility assessment, and final inclusion. A multistage screening strategy was employed to ensure that only studies relevant to international spread analysis, network modeling, data mining, machine learning, and spatiotemporal aspects of COVID-19 transmission. Figure 1 (Systematic Screening Processes for Paper Selection) shows the identification and screening stages, summarizes the eligibility assessment and data extraction workflow.

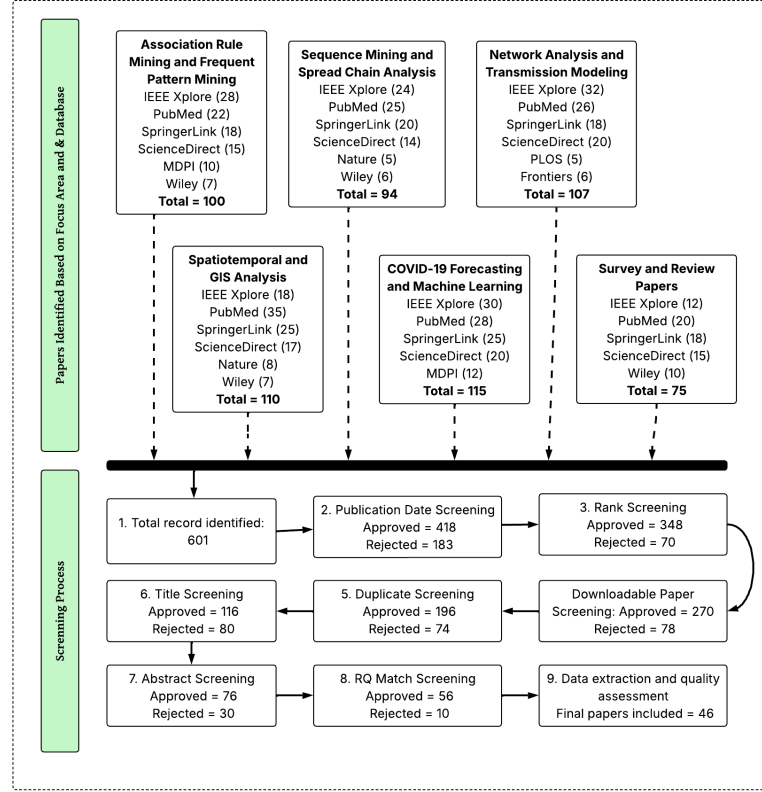


Fig. 1: Systematic Screening Processes For Paper Selection

3.1 Search Strategy and Information Sources

The literature search was conducted between 2021 and 2025 utilizing major scientific databases and online libraries, like IEEE Xplore, PubMed, Elsevier, Springer Nature, PLOS, BMC/Springer, Wiley Online Library, and Oxford University Press. Further references were identified through back and forward citation checking and through the initial spreadsheet. The search strategy contains keywords related to COVID-19, sequence mining, association rule mining, network analysis, transmission chains, and spatial spread. Peer-reviewed journal articles, high-quality surveys, and papers were prioritized. Also, citations were used to identify additional influential papers.

3.2 Eligibility Criteria

Eligibility criteria were defined based on the research objectives and research questions. The population of interest consisted of COVID-19 transmission systems at the patient, city, regional, or international scales. Interventions corresponded to computational approaches like association rule mining, machine learning, spatiotemporal

analysis, and network modeling. Outcomes included spread patterns, prediction performance, transmission pathways, and interpretable relationships. Empirical studies, methodological papers, and review papers were considered.

3.2.1 Inclusion Criteria

Studies were included the following criteria: Focused on COVID-19 or SARS-CoV-2 transmission, spread, prediction, or associated risk factors. Applied association rule mining, sequence analysis, network analysis, machine learning, GIS, or related computational methods. Reported sufficient methodological details, datasets, and experimental outcomes. Provided evidence relevant to at least one research question, written in English, was published between 2021 and 2025 in peer-reviewed journals or high-quality review articles.

3.2.2 Exclusion Criteria

Studies were excluded when they were unrelated to COVID-19 or infectious disease analytics. Focused exclusively on clinical treatment without methodological relevance. Lacked experimental or methodological details. Were duplicate publications. Were editorials, short notes, theses, or non-English papers. Could not be mapped to any research question. Did not provide sufficient information for data extraction.

3.3 Screening Processes

The screening process starts with 601 records and ended with 46 high quality papers, composed of seven stages, followed the PRISMA 2020 workflow. Zotero was used for reference management and duplicate detection and two reviewers independently evaluated papers throughout the process.

3.3.1 Publication Date Screening

The review focused on articles published between 2021 and 2025 because COVID-19 appeared in late 2019 and most key methodological developments occurred during this era. In this parts 418 papers approved and 183 out-of-scope works were rejected unless they presented essential algorithmic concepts.

3.3.2 Rank Screening

Studies passing abstract screening were ranked according to: Relevance to the research questions, Quality of publication venue, and Methodological completeness. Papers were subsequently classified into three relevance categories: Core papers, Supporting papers, Background papers. Only 348 medium- and high-relevance studies progressed to full-text review and 70 papers were rejected.

3.3.3 Downloadable Paper Screening

The downloadable paper screening phase keeps only accessible studies for further review. Papers without downloadable versions were excluded. In the outcome, 270 papers were accepted, and 78 papers were removed.

3.3.4 Duplication Screening

Duplicate records were discovered using DOI matching, PMID comparison, metadata similarity, and fuzzy title matching. When multiple versions of the same study existed, the most complete peer-reviewed and 196 papers was retained. 74 Conference papers superseded by journal versions were removed.

3.3.5 Title Screening

Titles were screened to determine relevance to COVID-19 analytics, association rule mining, sequence analysis, transmission networks, mobility analysis, or spatial spread. In this process 116 relevant papers were approved while 80 paper were rejected.

3.3.6 Abstract Screening

Abstracts were examined with 76 papers respect to: Problem formulation, Dataset characteristics, Analytical approach, experimental outcomes, and Relevance to the three research issues. 30 Studies without methodological evidence or relevant outcomes were excluded.

3.3.7 RQ Match Screening

Each paper was mapped to one or more research questions using a coding framework. RQ1: Travel-origin countries associated with international spread. Acceptable evidence, Mobility studies, Origin-destination analysis, International spread investigations. RQ2: Temporal, spatial, and network transmission patterns. GIS studies, Spatiotemporal analyses, Network reconstruction papers. RQ3: Interpretable prediction of spread routes, Association rule mining, Sequence mining, Forecasting, and transmission-chain studies. In this part 56 relevant papers were approved and 10 papers were excluded.

3.4 Data Extraction

A standardized extraction form was established and pilot-tested before full extraction. The extraction schema included: Study metadata (title, authors, year, venue), Dataset characteristics, Problem formulation, Preprocessing techniques, Algorithms and baselines, Evaluation metrics, Statistical tests, Key findings, Limitations, Software tools, Keywords and theoretical frameworks, Research question mapping, Relevance level. Qualitative synthesis was performed to identify methodological trends, strengths, limitations, and research gaps. Quantitative synthesis included publication trends, database distributions, thematic classifications, and research-question mappings. Quality assessment considered methodological rigor, reproducibility, dataset transparency, and relevance to international spread-chain analysis.

4 Results

The results section represents the key findings of the review paper by illustrating diagrams, bar charts, pie charts, maps, and tables, which Provide a clear overview of the papers that were selected and their main characteristics.

4.1 Paper Elimination Summary

Figure 2 shows the paper elimination process across seven screening stages. It starts with initial 601 papers, in publication data screening 183 papers, followed by quality 70 papers were removed, in downloading paper screening 78, duplicate screening 74, title screening 80, abstract screening 30, and RQ match screening 10 paper were rejected. After filtering stages, 46 unique papers remained for the final data extraction step.

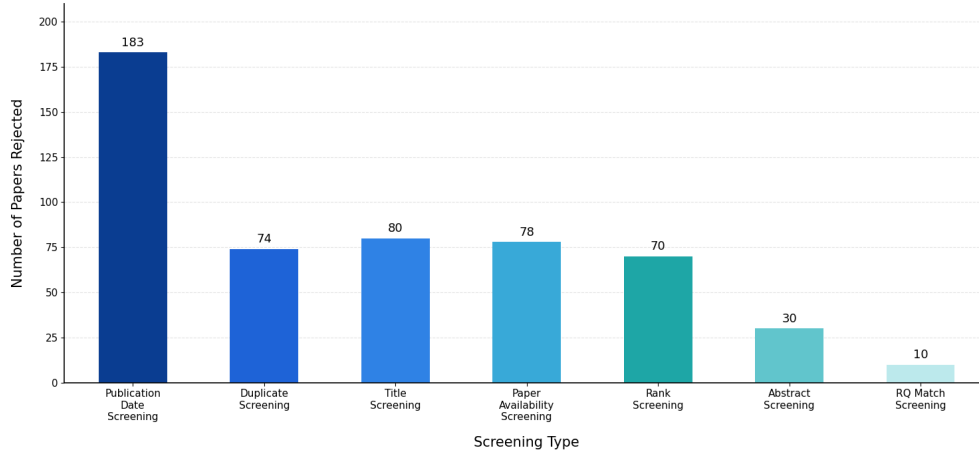


Fig. 2: Paper Elimination Process Breakdown.

4.1.1 Publication Trends Over the Years

Figure 3 shows publication counts from 2022 to 2024, starting from 2 papers in 2021 to a peak of 22 papers in 2022. The trend also illustrates 11 papers in 2023 and 11 papers in 2024. Generally, the trend shows an up down trajectory as years pass.

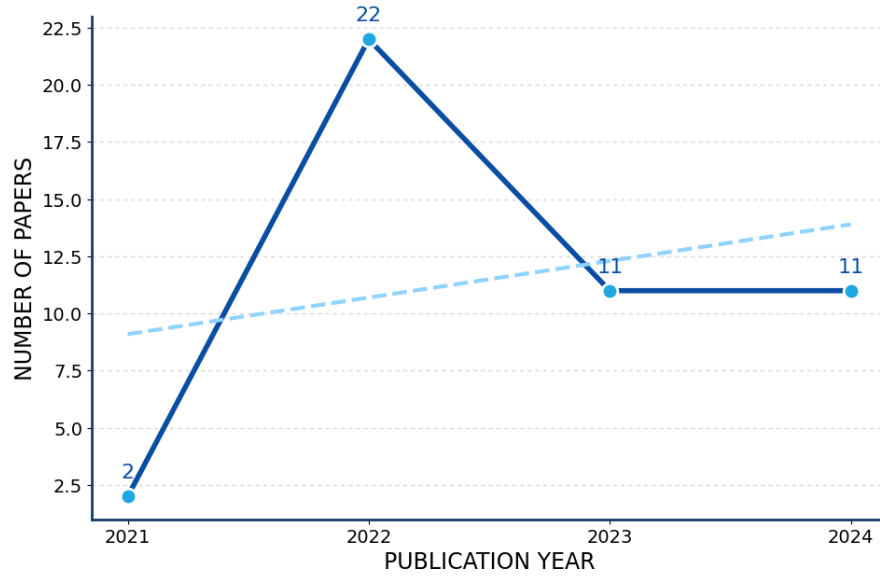


Fig. 3: Publication Trends Over the Years

4.1.2 Distribution of Papers by Database

The database distribution in Figure 4 demonstrate IEEE Xplore as the dominant source with 12 publications, followed by 10 papers in PubMed, and 8 papers in SpringerLink Digital Library. Respectively, 7, 5, 2, 1, and 1 papers were contributed by ScienceDirect, MDPI, Nature, PLOS and Wiley Online Library.

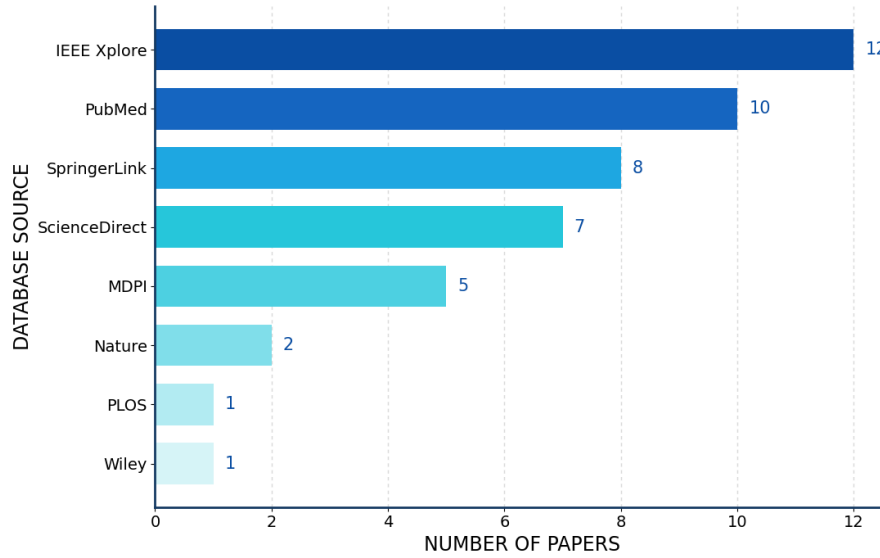


Fig. 4: Distribution of Research Papers by Database.

4.1.3 Geographical Distribution of Research Publications

The authors of the papers are from different countries, indicating global interests. The major hubs are the USA, China, and UK, and small hubs from places like Japan, France, Italy, and other countries, as shown in Figure 5.

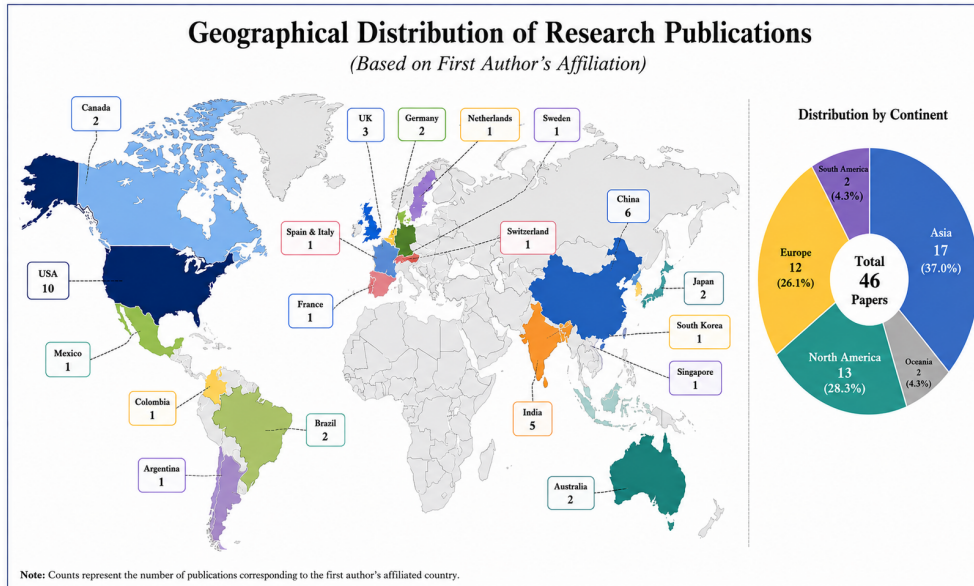


Fig. 5: Geographical Distribution of Research Publications.

4.1.4 Research Publication by Focus Area

To determine the direction of current research, papers are arranged according to particular subjects. The results in Figure 6 show a strong focus on GIS and spatiotemporal analysis. Additionally, the findings demonstrate the clear relationships between elements like modeling and forecasting, network and transmission, and Review with Survey studies.

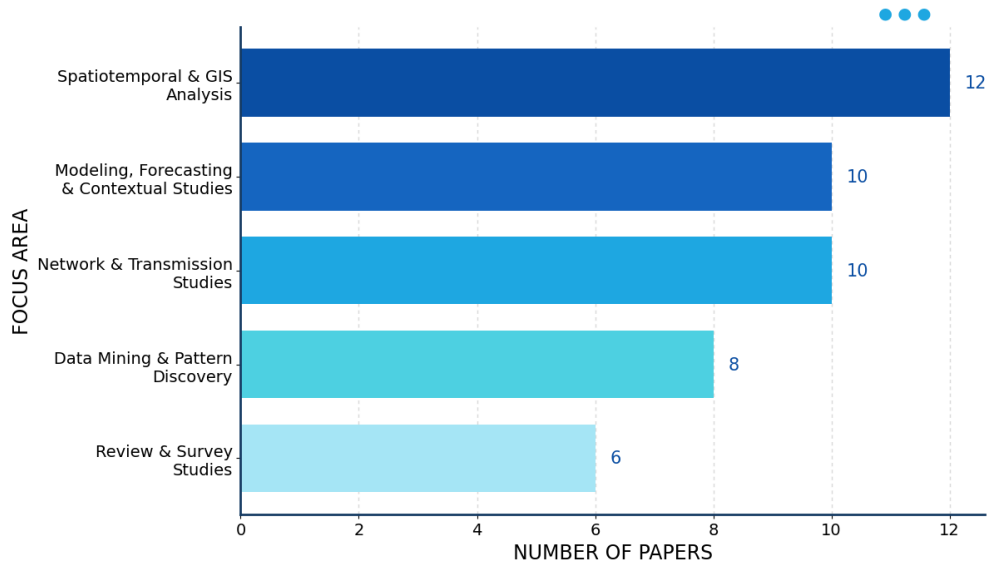


Fig. 6: Research Publication by Focus Area.

4.2 Methodological Approaches: Research Strategies and Techniques

We examined the research styles and data sources of our final papers to see how they investigated data mining performance. While most studies relied on Spatiotemporal, some studies discuss about Statistical modeling, Network analysis and Machine learning as shown in Figure 7 and 8.

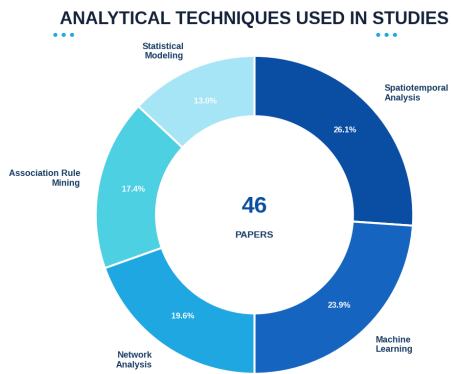


Fig. 7: Analytical Techniques Used in Studies

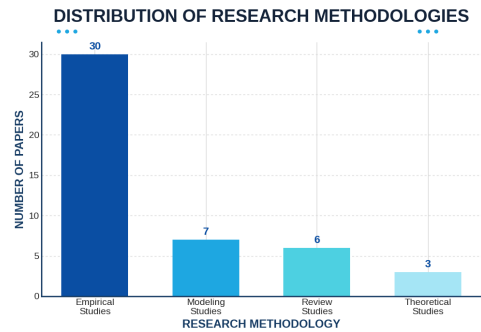


Fig. 8: Distribution of Research Methodologies

4.3 Cutting-Edge Technologies:

Figure 8 shows the technologies that were used in the research. It illustrates 26% of GIS and Spatiotemporal following by 24% of AI and Machine learning, 22% of Network and graph analytics, Datamining and pattern discovery with Statistical and forecasting tools have been used in this paper.

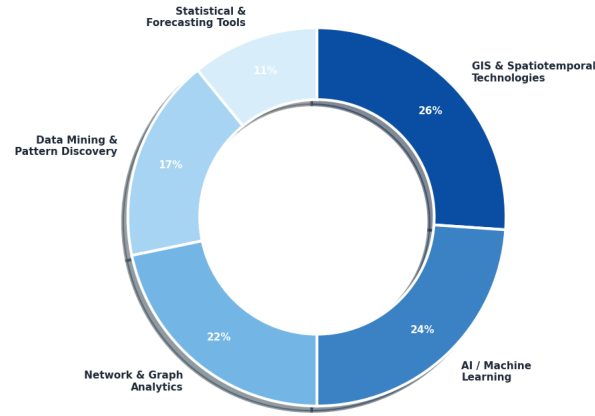


Fig. 9: Technologies Used in Research.

4.4 Breakthrough Findings: Key Outcomes from Selected Studies

The review shows several studies on research topics such as international spread and transmission pathways, spatiotemporal spread patterns, forecasting and prediction, network based transmission modeling, pattern discovery, and contact tracing privacy and surveillance as illustrated in Figure 10.

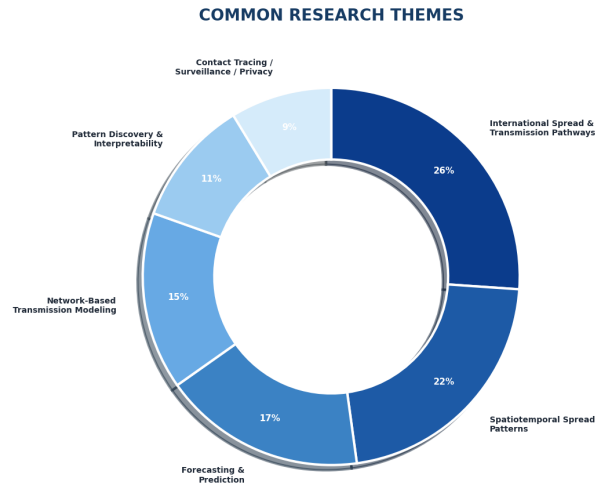


Fig. 10: Common Research Themes.

4.5 Applications and Real-World Relevance

Figure 10a shows that 39% of the studies discussed integration, while 61% does not clearly discussed. Figure 10b shows that reviewed studies mainly targeted researchers and data scientist. public health, epidemiologists, polict makers, healthcare surveillance and general public trancel risk.

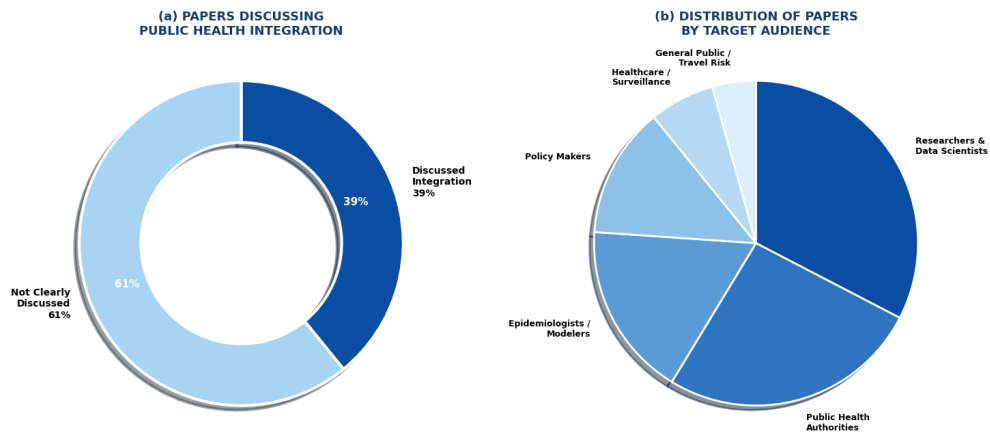


Fig. 11a: Paper Discussing Public Education Integration.

Fig. 11b: Distribution of Papers by Target Audience

4.6 Barriers to Implementation: Challenges and Limitations

Figure 12 shows algorithm complexity and interpretability as the main technical chal lenge among review studies, with additional emphasize to limited global datasets.

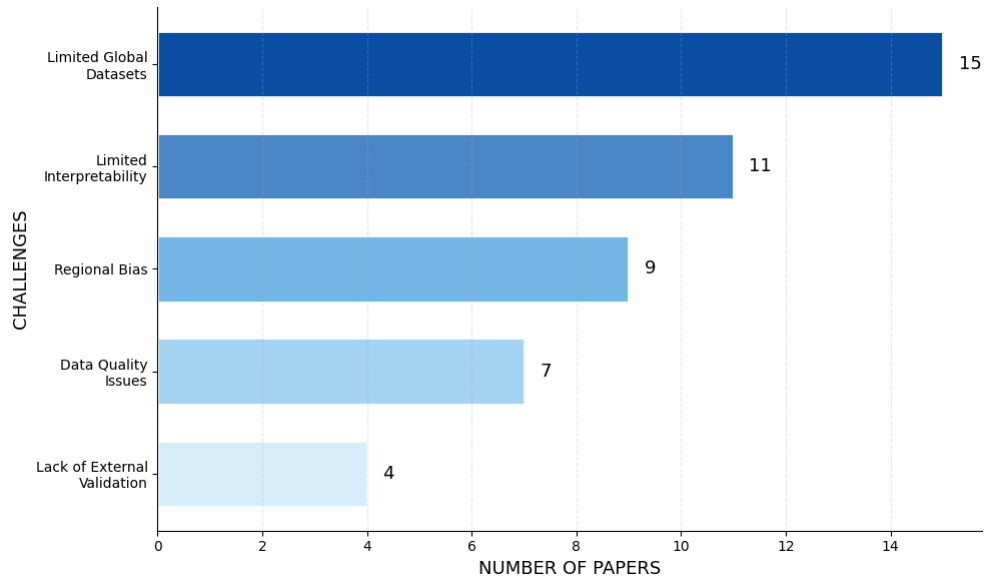


Fig. 12: Top Technical Challenges Identified.

4.7 Future Prospects: Research Gaps and Emerging Directions

Figure 13 visualizes the future research focuses; the findings indicate interpretable approaches, global datasets, hybrid frameworks, and cross country validation.

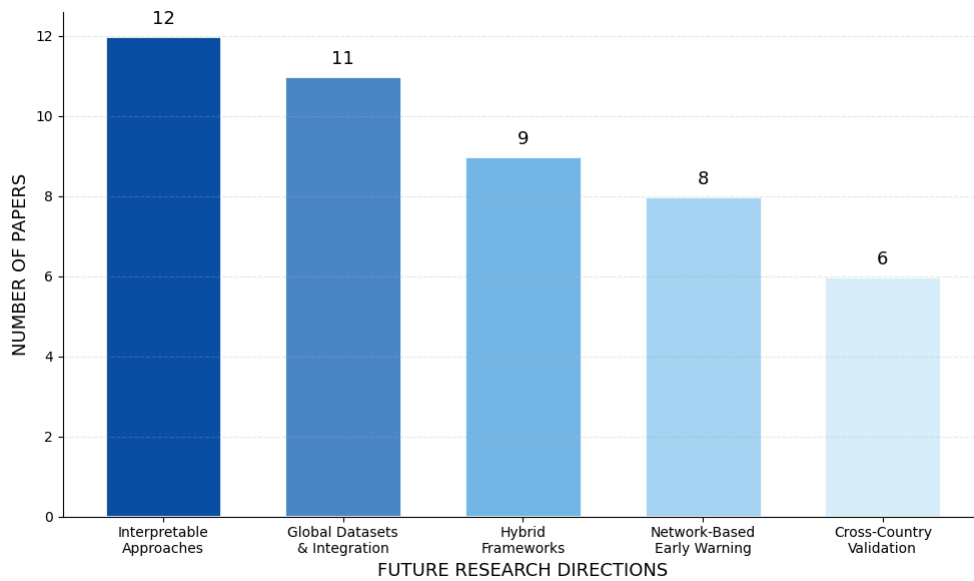


Fig. 13: Future Research Direction Proposed In Papers.

4.8 Impact on Society: Practical and Social Implications

Figure 14 shows that policy implications receive the least attention; fewer studies focus on social impact. In contrast, practical impact followed by theoretical impact obtain higher attention.

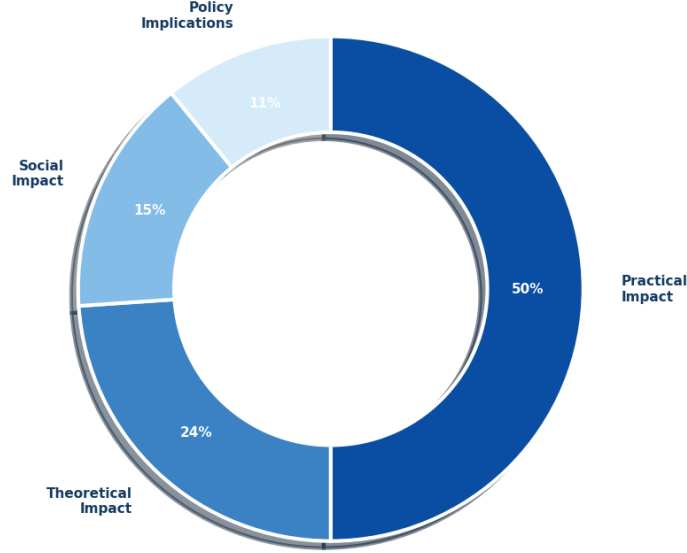


Fig.14:Breakdown of the Impact Made by the Research Papers.

4.9 Contribution and Research Questions Matching

This section explains how the selected 46 studies contribute to COVID-19 international spread analysis and how their contributions map to the predefined research questions. Every paper was coded according to its main contribution type and its relevance to RQ1, RQ2, and RQ3. The coding scheme considered six contribution categories: data mining and pattern discovery, network and transmission modeling, spatiotemporal and GIS analysis, forecasting and machine learning, contextual and environmental analysis, and review/survey synthesis. The mapping rules were defined as follows. A paper was marked as fully answered if it directly addressed a research question through empirical results, explicit methodology, or strong analytical evidence. A paper was marked as partially answered if it provided supporting concepts, transferable methods, or contextual evidence without directly investigating the exact question. A paper was marked as not answered if its contribution was outside the scope of that research question. For RQ1, papers were considered relevant if they discussed travel-origin countries, mobility patterns, origin–destination relationships, international spread, cross-border transmission, or source–location characteristics. For RQ2, papers were mapped if they analyzed temporal, spatial, GIS-based, network, or transmission-pattern evidence. For

RQ3, papers were mapped if they contributed analytical methods or computational approaches capable of providing interpretable evidence regarding future spread routes or high-risk transmission pathways. The RQ mapping showed that RQ2 received the strongest coverage, with 33 papers contributing to temporal, spatial, or network transmission patterns. RQ3 was supported by 37 papers through data mining, machine learning, network analysis, forecasting, and interpretable modeling approaches. RQ1 was addressed by 25 papers, primarily through mobility, spatiotemporal, and transmission studies. This distribution indicates that the literature strongly supports the analysis of spatial and network patterns, whereas direct evidence concerning international origin–destination relationships remains comparatively less developed. Overall, the selected papers demonstrate considerable methodological diversity, but their contributions are unevenly distributed across the research questions. Most studies focus on prediction, spatial clustering, local transmission, or case-count modeling. Comparatively fewer studies directly investigate international transmission mechanisms and origin–destination relationships among countries. This confirms the principal research gap addressed by the present study.

4.9.1 Research Question Matching Analysis

The table examines how effectively each selected study contributes to solving the identified research questions (RQ1, RQ2, and RQ3). Each study categorized either fully answered, partially answered, or not answered based on the amount to which its findings, techniques, or evidence address the precise aims of the research questions.

Table 1: Analysis of Research Questions Matching

Research Question	Fully Answered	Partially Answered	Not Answered
RQ1: Which travel-origin countries or regions are most frequently associated with international first-case introductions?	8	17	21
RQ2: What temporal, spatial, or network transmission patterns emerge among countries from rapid spread evidence?	15	18	13
RQ3: What analytical methods and computational approaches provide interpretable evidence for potential future spread routes or high-risk pathways?	15	22	9

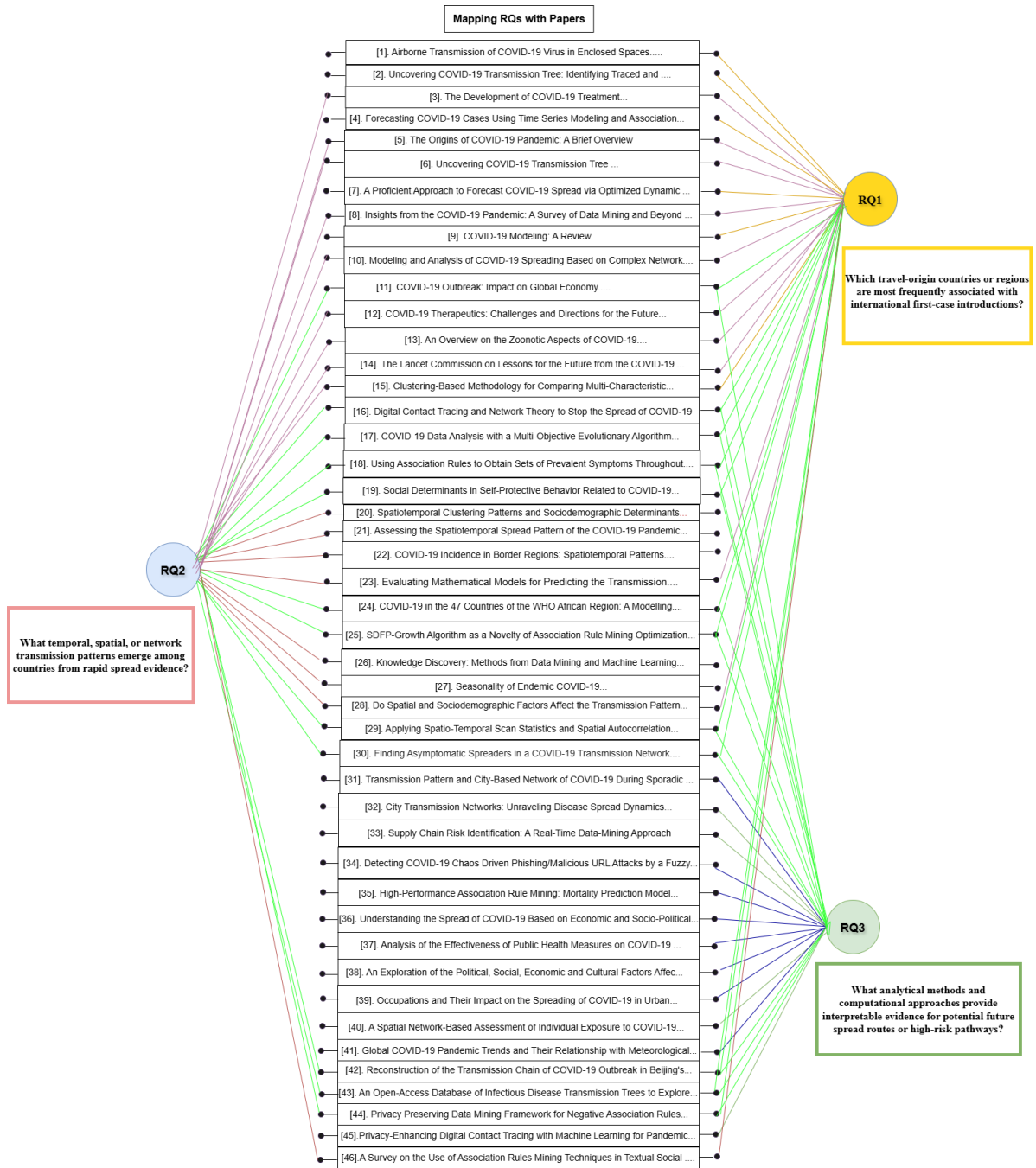


Fig. 2: Breakdown of the Impact Made by the Number of Research Papers.

5 Comparison and Analysis

Table 2: Comparison with Related Review Studies

Work	Scope	Key Characteristics	Results
This Work	COVID-19 spread-chain analysis using association rule mining, sequence representations, spatiotemporal analysis, and transmission networks	PRISMA-based SLR with 46 papers; focuses on international first-case introductions; maps studies to RQ1–RQ3; emphasizes interpretability and spread pathways	Identified methodological trends, research gaps, and the potential of ARM and sequence-based approaches for discovering high-risk international spread routes.
Afyouni et al. (2024) [8]	General COVID-19 machine learning and forecasting techniques	Broad survey of machine learning, deep learning, and statistical models; prediction-oriented	Concluded that machine learning models were widely used but highlighted issues of generalizability and explainability.
Serafino et al. (2022)[16]	Data mining and analytical approaches for COVID-19	Covers machine learning, time series, and network analysis methods	Found that most studies focused on prediction accuracy rather than interpretable knowledge discovery.
Liu et al. (2022)[41]	Digital contact tracing and privacy-preserving analytics	Reviews implementation frameworks and privacy concerns	Demonstrated trade-offs between data utility, scalability, and privacy preservation.
Diaz-Garcia et al. (2022)[46]	ARM applications in social media and textual data	Focuses on interpretable co-occurrence discovery using rule mining	Showed that ARM provides explainable knowledge extraction from large datasets but lacks application to international transmission chains.
Alali et al. (2022)[7]	Pandemic modeling and forecasting studies	Summarizes environmental, socioeconomic, and predictive factors	Highlights the importance of contextual variables but does not investigate origin–destination spread chains.

6 Discussion

This literature study combined 46 high quality works pertinent to COVID-19 international spread analysis, including data mining, network science, machine learning, and spatiotemporal analytics. The findings revealed that the majority of the studies focused on predicting, geographical clustering, and local transmission dynamics, making RQ2 the most highly supported research topic. In contrast, fewer research directly studies international transmission mechanisms and origin–destination linkages among countries. Another key finding is that existing research depends mostly on regional and incidence-based information, while globally integrated transmission data remain sparse. Moreover, many study stress prediction performance above interpretability and suffer from minimal external validation. Although several computational approaches

have been applied, they are often investigated independently rather than inside integrated frameworks. Overall, the research provides substantial evidence for temporal, geographical, and network transmission patterns but comparatively less support for travel-origin links and future spread paths. These findings underscore the need for internationally representative datasets, interpretable analytical methodologies, and improved validation to increase knowledge of worldwide transmission patterns and strengthen preparedness for future pandemics.

6.1 Key Findings and Implications

The systematic review of 46 selected papers reveal significant trends in COVID-19 global spread research. Firstly the paper is dominated by papers focusing on temporal, spatial, and network transmission patterns, revealing researchers had considerable emphasis on the propagation of disease across, population, and regions. The main analytical tools are, network modeling, machine learning, and Spatiotemporal analysis, indicating the growing reliance on computational methods for pandemic analysis. Secondly, Although numerous studies explore prediction and dynamic transmission fewer studies examine travel-origin relationships and first case introduction among the countries. Moreover, most analyses are on mobility information, local transmission records and regional data, rather than a global spread chain dataset. This reveals the limited understanding of the global pathway of COVID-19 during early stages of transmission. Third, The paper illustrates that selected studies use various datasets consisting epidemiological records, mobility datasets, transmission trees, clinical records, environmental variables, and social media data. However the variation in data quality, reporting standards and geographical coverage makes the cross-study comparison difficult. Finally, The review demonstrates that while forecasting accuracy is the main objective of many papers, less attention has been given to analytical approaches that can examine how and why global spread happens. These findings indicate that future research could balance predictive performance by adding explanation and boost preparedness and decision making for pandemics.

6.2 Challenges and Limitations

Beside all the findings, the review consists of some limitations, firstly although the review followed a PRISMA-based screening process, it is limited to 46 papers based on the mentioned inclusion and exclusion criteria. Additionally, the relevant studies are published outside the selected database. Second, the reviewed studies show a lot of heterogeneity, in dataset choice analytical methods evaluation metrics, and also how results are reported. This variety prevents a straightforward numeric comparison between papers as a result the whole review mostly leans toward a qualitative synthesis rather than doing a meta-analysis. Third, many of the included studies rely on regional or country-specific datasets instead of globally representative international transmission data. Differences in surveillance systems, reporting accuracy, and data availability may affect the generalizability of the reported findings. Third, a main part of the included studies lean on regional, or even country specific datasets rather than on internationally representative transmission data. variations in surveillance systems,

how accurate the reporting is, and what data is even available may mess with how well the reported findings generalize to other settings.

Finally, putting the studies into thematic groups and mapping them to the research questions kind of inevitably brings in a bit of expert judgment. Even though the same coding framework was used across the whole review, some of the studies turned up in more than one research question, or even across different method categories, so that could lead to small subjectivity in the synthesis.

7 Conclusion

This study offered a PRISMA-based systematic literature evaluation of 46 papers relevant to COVID-19 international spread analysis, data mining, network science, machine learning, and spatiotemporal modeling. The review discovered that current research is mostly dominated by forecasting models, machine learning methodologies, and regional transmission investigations, whereas comparably fewer studies investigate international transmission mechanisms and origin–destination links among countries. The findings showed that temporal, spatial, and network transmission patterns are well explored, while identifying travel-origin countries and understanding potential future spread pathways remain comparatively under-addressed. The reviewed study suggested that a wide range of analytical tools, including data mining, network analysis, forecasting approaches, and spatiotemporal methodologies, provide useful information on disease transmission. However, the literature also revealed substantial weaknesses, including reliance on regional datasets, inadequate external validation, diversity in reporting systems, and insufficient interpretability in analytical frameworks. In particular, integrating several computational methodologies with international transmission evidence can provide deeper insight into high-risk pathways and lead to the creation of more effective early warning systems for future infectious disease epidemics. Future studies should emphasize explainability, global datasets, and interdisciplinary approaches to improve understanding of international transmission dynamics and improve pandemic preparedness.

References

- [1] X. Zhao, S. Liu, Y. Yin, T. Zhang, and Q. Chen, “Airborne transmission of covid-19 virus in enclosed spaces: an overview of research methods,” *Indoor air*, vol. 32, no. 6, p. e13056, 2022.
- [2] H. Lee, H. Choi, H. Lee, S. Lee, and C. Kim, “Uncovering covid-19 transmission tree: identifying traced and untraced infections in an infection network,” *Frontiers in Public Health*, vol. 12, p. 1362823, 2024.
- [3] Y. Yuan, B. Jiao, L. Qu, D. Yang, and R. Liu, “The development of covid-19 treatment,” *Frontiers in immunology*, vol. 14, p. 1125246, 2023.
- [4] R. Somyanonthanakul, K. Warin, W. Amasiri, K. Mairiang, C. Mingmalairak, W. Panichkitkosolkul, K. Silanun, T. Theeramunkong, S. Nitikraipot, and S. Suebnukarn, “Forecasting covid-19 cases using time series modeling and association rule mining,” *BMC medical research methodology*, vol. 22, no. 1, p. 281, 2022.
- [5] Y.-J. Hao, Y.-L. Wang, M.-Y. Wang, L. Zhou, J.-Y. Shi, J.-M. Cao, and D.-P. Wang, “The origins of covid-19 pandemic: A brief overview,” *Transboundary and emerging diseases*, vol. 69, no. 6, pp. 3181–3197, 2022.
- [6] H. Lee, H. Choi, S. Lee, C. Lee, and C. Kim, “Uncovering covid-19 transmission tree: identifying traced and untraced infections in an infection network,” *Frontiers in Public Health*, vol. 12, 2024.
- [7] Y. Alali, F. Harrou, and Y. Sun, “A proficient approach to forecast covid-19 spread via optimized dynamic machine learning models,” *Scientific Reports*, vol. 12, no. 1, p. 2467, 2022.
- [8] I. Afyouni, I. Hashim, Z. Aghbari, T. Elsaka, M. Almahmoud, and L. Abualigah, “Insights from the COVID-19 pandemic: A survey of data mining and beyond,” *Applied Spatial Analysis and Policy*, vol. 17, pp. 1359–1411, 2024, published online: 22 June 2024. [Online]. Available: <https://doi.org/10.1007/s12061-024-09588-5>
- [9] L. Cao and Q. Liu, “COVID-19 modeling: A review,” *ACM Computing Surveys*, vol. 57, no. 1, pp. 1–42, 2024, article 10, October 2024. [Online]. Available: <https://doi.org/10.1145/3686150>
- [10] R. Wang, Y. Chen, Y. Liu, L. Ge, Y. Liu, and M. Tang, “Modeling and analysis of covid-19 spreading based on complex network theory,” *Europhysics Letters*, vol. 148, no. 1, p. 11001, 2024.
- [11] S. Naseer, S. Khalid, S. Parveen, K. Abbass, H. Song, and M. V. Achim, “Covid-19 outbreak: Impact on global economy,” *Frontiers in public health*, vol. 10, p. 1009393, 2023.
- [12] P. C. Robinson, D. F. Liew, H. L. Tanner, J. R. Grainger, R. A. Dwek, R. B. Reisler, L. Steinman, M. Feldmann, L.-P. Ho, T. Hussell *et al.*, “Covid-19 therapeutics: Challenges and directions for the future,” *Proceedings of the National Academy of Sciences*, vol. 119, no. 15, p. e2119893119, 2022.
- [13] A. S. Ray and K. Bhattacharya, “An overview on the zoonotic aspects of covid-19,” *Proceedings of the National Academy of Sciences, India Section B: Biological Sciences*, vol. 94, no. 1, pp. 9–13, 2024.

- [14] J. D. Sachs, S. S. A. Karim, L. Akin, J. Allen, K. Brosbøl, F. Colombo, G. C. Barron, M. F. Espinosa, V. Gaspar, A. Gaviria *et al.*, “The lancet commission on lessons for the future from the covid-19 pandemic,” *The Lancet*, vol. 400, no. 10359, pp. 1224–1280, 2022.
- [15] A. Kirpich, A. Shishkin, P. Lhewa, E. Adeniyi, M. Norris, G. Chowell, Y. Gankin, P. Skums, and A. Perez Tchernov, “Clustering-based methodology for comparing multi-characteristic epidemiological dynamics with application to COVID-19 epidemiology in Europe,” *Royal Society Open Science*, vol. 12, no. 250440, pp. 1–21, 2025, supplementary material available at <https://doi.org/10.6084/m9.figshare.c.8042223>. [Online]. Available: <https://doi.org/10.1098/rsos.250440>
- [16] M. Serafino, H. S. Monteiro, S. Luo, S. D. S. Reis, C. Igual, A. S. Lima Neto, M. Travizano, J. S. Andrade, and H. A. Makse, “Digital contact tracing and network theory to stop the spread of COVID-19 using big-data on human mobility geolocalization,” *PLOS Computational Biology*, vol. 18, no. 4, p. e1009865, 2022, published: April 11, 2022. [Online]. Available: <https://doi.org/10.1371/journal.pcbi.1009865>
- [17] S. Sinisterra-Sierra, S. Godoy-Calderon, and M. Pescador-Rojas, “COVID-19 data analysis with a multi-objective evolutionary algorithm for causal association rule mining,” *Mathematical and Computational Applications*, vol. 28, no. 1, p. 12, 2023, published: 13 January 2023. [Online]. Available: <https://doi.org/10.3390/mca28010012>
- [18] J. G. Marques, B. M. d. Carvalho, L. A. Guedes, and M. Da Costa-Abreu, “Using association rules to obtain sets of prevalent symptoms throughout the covid-19 pandemic: an analysis of similarities between cases of covid-19 and unspecified sars in são paulo-brazil,” *International Journal of Environmental Research and Public Health*, vol. 21, no. 9, p. 1164, 2024.
- [19] G. Urbanin, W. Meira, A. Serpa, D. de Souza Costa, L. Baldaçara, A. P. da Silva, R. Guatimosim, A. M. Lacerda, E. A. Oliveira, A. Braule *et al.*, “Social determinants in self-protective behavior related to covid-19: Association rule-mining study,” *JMIR Public Health and Surveillance*, vol. 8, no. 6, p. e34020, 2022.
- [20] M. Sijlander, R. Uusitalo, P. Pellikka, S. Isosomppi, and O. Valpolahti, “Spatiotemporal clustering patterns and sociodemographic determinants of covid-19 (sars-cov-2) infections in helsinki, finland,” *Spatial and Spatio-temporal Epidemiology*, vol. 41, p. 100493, 2022.
- [21] Y. L. Leong, S. M. Ghazali, M. K. Che Ibrahim *et al.*, “Assessing the spatiotemporal spread pattern of the covid-19 pandemic in malaysia,” *Frontiers in Public Health*, vol. 10, p. 836358, 2022.
- [22] T. Chilla, T. Gröbe, S. Hippe, and B. B. Walker, “Covid-19 incidence in border regions: spatiotemporal patterns and border control measures,” *Public Health*, vol. 202, pp. 80–83, 2022.
- [23] A. K. Sabherwal, A. Sood, and M. A. Shah, “Evaluating mathematical models for predicting the transmission of covid-19 and its variants towards sustainable health and well-being,” *Discover Sustainability*, vol. 5, no. 1, p. 38, 2024.
- [24] J. W. Cabore, H. C. Karamagi, H. K. Kipruto, J. K. Mungatu, J. A. Asamani,

- B. Droti, R. Titi-Ofei, A. B. W. Seydi, S. N. Kidane, T. Balde *et al.*, “Covid-19 in the 47 countries of the who african region: a modelling analysis of past trends and future patterns,” *The lancet global health*, vol. 10, no. 8, pp. e1099–e1114, 2022.
- [25] B. Siswanto, H. Soeparno, N. F. Sianipar, and W. Budiharto, “Sdfp-growth algorithm as a novelty of association rule mining optimization,” *IEEE access*, vol. 12, pp. 21 491–21 502, 2024.
- [26] X. Shu and Y. Ye, “Knowledge discovery: Methods from data mining and machine learning,” *Social Science Research*, vol. 110, p. 102817, 2023.
- [27] J. P. Townsend, H. B. Hassler, A. D. Lamb, P. Sah, A. Alvarez Nishio, C. Nguyen, A. D. Tew, A. P. Galvani, and A. Dornburg, “Seasonality of endemic covid-19,” *MBio*, vol. 14, no. 6, pp. e01 426–23, 2023.
- [28] S. Fortuna, R. P. Setiawan, and A. Sharifi, “Do spatial and sociodemographic factors affect the transmission pattern of covid-19? evidence from surabaya city, indonesia,” *International Journal of Disaster Risk Reduction*, vol. 96, p. 103900, 2023.
- [29] L. R. d. A. Morais and G. S. d. S. Gomes, “Applying spatio-temporal scan statistics and spatial autocorrelation statistics to identify covid-19 clusters in the world – a vaccination strategy?” *Spatial and Spatio-temporal Epidemiology*, vol. 39, p. 100461, 2021.
- [30] Z. Liu, Y. Ma, Q. Cheng, and Z. Liu, “Finding asymptomatic spreaders in a covid-19 transmission network by graph attention networks,” *Viruses*, vol. 14, no. 8, p. 1659, 2022.
- [31] L. Zhao, X. Xu, J. Jiang, and D. Wang, “Transmission pattern and city-based network of covid-19 during sporadic outbreaks,” *Cities*, vol. 154, p. 105326, 2024.
- [32] H. Alrasheed, N. Alballa, I. Al-Turaiki, F. Almutlaq, and R. Aljulaid, “City transmission networks: Unraveling disease spread dynamics,” *ISPRS International Journal of Geo-Information*, vol. 13, no. 8, p. 283, 2024.
- [33] A. D. Ganesh and P. Kalpana, “Supply chain risk identification: A real-time data-mining approach,” *Industrial Management & Data Systems*, vol. 122, no. 5, pp. 1333–1354, 2022.
- [34] S. R. Zahra, M. A. Chishti, A. I. Baba, and F. Wu, “Detecting covid-19 chaos driven phishing/malicious url attacks by a fuzzy logic and data mining based intelligence system,” *Ain Shams Engineering Journal*, 2021.
- [35] R. G. Nadakinamani, A. Reyana, Y. Gupta, S. Kautish, S. Ghorashi, M. M. Jamjoom, and A. W. Mohamed, “High-performance association rule mining: Mortality prediction model for cardiovascular patients with covid-19 patterns,” *Alexandria Engineering Journal*, vol. 71, pp. 347–354, 2023.
- [36] L. Dinia, V. A. Iannitti, F. Mangini, F. Di Lascio, and F. Frezza, “Understanding the spread of covid-19 based on economic and socio-political factors,” *Sustainability*, vol. 14, no. 3, p. 1768, 2022.
- [37] T. C. Silva, L. Anghinoni, C. P. d. Chagas, L. Zhao, and B. M. Tabak, “Analysis of the effectiveness of public health measures on covid-19 transmission,” *International Journal of Environmental Research and Public Health*, vol. 20, no. 18, p. 6758, 2023.
- [38] J. W. Tang, M. A. Caniza, M. Dinn, D. E. Dwyer, J.-M. Heraud, L. C. Jennings,

- J. Kok, K. O. Kwok, Y. Li, T. P. Loh *et al.*, “An exploration of the political, social, economic and cultural factors affecting how different global regions initially reacted to the covid-19 pandemic,” *Interface focus*, vol. 12, no. 2, p. 20210079, 2022.
- [39] M.-G. Hâncean, J. Lerner, M. Perc, I. Oană, D.-A. Bunaciu, A. A. Stoica, and M.-C. Ghiță, “Occupations and their impact on the spreading of covid-19 in urban communities,” *Scientific Reports*, vol. 12, p. 14115, 2022.
 - [40] Z. Kan, M.-P. Kwan, J. Huang, J. Cai, and D. Liu, “A spatial network-based assessment of individual exposure to covid-19,” *Annals of the American Association of Geographers*, vol. 114, no. 8, pp. 1693–1703, 2024.
 - [41] H. Liu *et al.*, “Global covid-19 pandemic trends and their relationship with meteorological variables, air pollutants and socioeconomic aspects,” *Environmental Research*, vol. 204, p. 112249, 2022.
 - [42] T. Luo, J. Wang, Q. Wang, X. Wang, P. Zhao, D. D. Zeng, Q. Zhang, and Z. Cao, “Reconstruction of the transmission chain of covid-19 outbreak in beijing’s xinfadi market, china,” *International Journal of Infectious Diseases*, vol. 116, pp. 411–417, 2022.
 - [43] J. C. Taube, P. B. Miller, and J. M. Drake, “An open-access database of infectious disease transmission trees to explore superspreader epidemiology,” *PLOS Biology*, vol. 20, no. 6, p. e3001685, 2022.
 - [44] S. M. Darwish, R. M. Essa, M. A. Osman, and A. A. Ismail, “Privacy preserving data mining framework for negative association rules: An application to healthcare informatics,” *IEEE Access*, vol. 10, pp. 76 268–76 280, 2022.
 - [45] C.-N. Hang, Y.-Z. Tsai, P.-D. Yu, J. Chen, and C.-W. Tan, “Privacy-enhancing digital contact tracing with machine learning for pandemic response: A comprehensive review,” *Big Data and Cognitive Computing*, vol. 7, no. 2, p. 108, 2023, published: 1 June 2023. [Online]. Available: <https://doi.org/10.3390/bdcc7020108>
 - [46] V. Authors, “A survey on the use of association rules mining techniques in textual social media,” *Artificial Intelligence Review*, vol. 56, pp. 1175–1200, 2023.